

Evaluating Personality Expression in Large Language Models

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Abstract

Large language models are increasingly used to produce text on behalf of users or synthetic personas, yet it is unclear whether they can faithfully express personality expression in unconstrained writing. We address this on the Pennebaker & King stream-of-consciousness Essays dataset by training a multi-label RoBERTa probe to score trait expression in text, generating essays with GPT-4o-mini under single-trait (Style B) and full-profile (Style A) prompting, and comparing predicted trait distributions and per-essay alignment against human baselines. We further analyse linguistic cues via LIWC-style features. This paper describes the end-to-end system: data preparation, probe training, async LLM generation, distributional evaluation, and qualitative error analysis.

1. Introduction

Large language models can produce fluent stream-of-consciousness text, but it remains unclear whether they can reliably adopt a requested personality the way people do when they write freely. In this work we ask a concrete version of that question: if we instruct GPT-4o-mini to write as someone who is, for example, highly extraverted or very anxious, does the resulting essay look—to an automatic personality scorer trained on human writing—like a human essay with that trait, or does the model only mimic obvious wording from the prompt? We study this on the Pennebaker & King Essays dataset, where students wrote unstructured essays and received binary Big Five labels, and we train RoBERTa-base as a multi-label *probe* that predicts those five traits from text; the probe is our measurement instrument, not a claim about ground-truth personality. We then generate new essays in two regimes—Style B varies a single trait at a time (HIGH, LOW, or no trait instruction) to test whether steering one dimension works in isolation, while Style A copies full five-trait profiles from real human annotators to test more realistic multi-trait control—and for every generated essay we run the probe and compare its outputs to the personality we asked for and to the distribution of scores on held-out human essays, using distributional distances and density plots for Style B, per-essay agreement for Style A, and LIWC-style linguistic analysis to inspect which cues differ between human and LLM text. This paper describes that end-to-end pipeline—data preparation, probe training, LLM generation, evaluation, and error analysis—and reports how faithfully explicit personality instructions translate into detectable stylistic shifts.

2. Second section

In scientific papers, this section usually (but not necessarily) briefly describes the related research and what makes the presented approach different from it.

2.1. First subsection

This is a subsection of the second section.



Figure 1: This is the figure caption. Full sentences should be followed with a dot. The caption should be placed *below* the figure. Caption should be short; details should be explained in the text.

2.2. Second subsection

This is the second subsection of the second section. Referencing the (sub)sections in text is performed as follows: “in Section ?? we have shown ...”.

2.2.1. Sub-subsection example

This is a sub-subsection. If possible, it is better to avoid sub-subsections.

3. Extent of the paper

The paper should have a minimum of 3 and a maximum of 4 pages, plus an additional page for references.

4. Figures and tables

4.1. Figures

Here is an example on how to include figures in the paper. Figures are included in $\LaTeX$ code immediately *after* the text in which these figures are referenced. Allow $\LaTeX$ to place the figure where it believes is best (usually on top of the page or at the position where you would not place the figure). Figures are referenced as follows: “Figure ?? shows ...”. Use tilde (~) to prevent separation between the word “Figure” and its enumeration.

Table 2: Wide-table caption

Heading1	Heading2	Heading3
A	A very long text, longer that the width of a single column	128
B	A very long text, longer that the width of a single column	3123
C	A very long text, longer that the width of a single column	−32

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